

BASIC SCIENCES

Histidine-Rich Glycoprotein Modulates the Toxic Effects of High-Dose Polyphosphate in Mice

Rida A. Malik^{ID}*, Ji Zhou^{ID}*, James C. Fredenburgh^{ID}, Jeff Crosby, Alexey S. Revenko^{ID}, Jeff S. Healey^{ID}, Jeffrey I. Weitz^{ID}

BACKGROUND: Polyphosphate (polyP), a procoagulant released from platelets, activates coagulation via the contact system and modulates cardiomyocyte viability. High-dose intravenous polyP is lethal in mice, presumably because of thrombosis. Previously, we showed that HRG (histidine-rich glycoprotein) binds polyP and attenuates its procoagulant effects. In this study, we investigated the mechanisms responsible for the lethality of intravenous polyP in mice and the impact of HRG on this process.

METHODS: The survival of wild-type or HRG-deficient mice given intravenous synthetic or platelet-derived polyP in doses up to 50 mg/kg or saline was compared. To determine the contribution of thrombosis, the effect of FXII (factor XII) knockdown or enoxaparin on polyP-induced fibrin deposition in the lungs was examined. To assess cardiotoxicity, the ECG was continuously monitored, the levels of troponin I and the myocardial band of creatine kinase were quantified, and the viability of a cultured murine cardiomyocyte cell line exposed to polyP in the absence or presence of HRG was determined.

RESULTS: In HRG-deficient mice, polyP was lethal at 30 mg/kg, whereas it was lethal in wild-type mice at 50 mg/kg. Although FXII knockdown or enoxaparin administration attenuated polyP-induced fibrin deposition in the lungs, neither affected mortality. PolyP induced dose-dependent ECG abnormalities, including heart block and ST-segment changes, and increased the levels of troponin and myocardial band of creatine kinase, effects that were more pronounced in HRG-deficient mice than in wild-type mice and were attenuated when HRG-deficient mice were given supplemental HRG. Consistent with its cardiotoxicity, polyP reduced the viability of cultured cardiomyocytes in a dose-dependent manner, an effect attenuated with supplemental HRG.

CONCLUSIONS: High-dose intravenous polyP is cardiotoxic in mice, and HRG modulates this effect.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: arrhythmias, cardiac ■ cardiotoxicity ■ factor XII ■ polyphosphates ■ thrombosis

Inorganic polyphosphates (polyPs) are linear polymers of phosphate residues linked by high-energy phosphoanhydride bonds.¹ Each phosphate residue carries a monovalent negative charge, rendering polyP highly anionic at physiological pH.² PolyP is ubiquitous and varies in polymer length from a few to several thousand phosphate units depending on the organism and the tissue in which it is synthesized.² In mammals, polyP has been implicated in blood coagulation, innate immunity, inflammation, cancer metastasis, cell proliferation, and cell death.^{3,4} Therefore, polyP has myriad functions.

In humans, large amounts of polyP are found in the dense granules of platelets.^{5,6} Platelet polyP, which is 60 to 100 phosphate units in length, is released when

platelets are activated.^{5,6} Platelet polyP promotes coagulation by inducing FXII (factor XII) autoactivation,^{7–10} promoting FV (factor V) and FXI (factor XI) activation by thrombin,^{11–13} and enhancing the stability of fibrin clots.^{14–16} Therefore, platelet polyP appears to be an important physiological activator of coagulation.

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PolyP also functions at the cellular level. PolyP is found in the heart, kidneys, and brain.^{17–19} In cardiac mitochondria, polyP activates the mitochondrial permeability transition pore, a nonselective pore in the inner mitochondrial

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Nonstandard Abbreviations and Acronyms	
ASO	antisense oligonucleotide
CK-MB	creatinine kinase myocardial band
DAPI	4',6-diamidino-2-phenylindole
FV	factor V
FXI	factor XI
FXII	factor XII
FXIIa	activated factor FXII
HEK	human embryonic kidney
HRG	histidine-rich glycoprotein
HRG^{-/-}	histidine-rich glycoprotein-deficient
LDH	lactate dehydrogenase
PolyP	polyphosphate
TAT	thrombin-antithrombin

membrane that provides a permeation pathway for ions, solutes, and small proteins.²⁰⁻²² By allowing their entry, polyP may modulate cardiomyocyte viability because polyP depletion protects cardiomyocytes from ischemic injury.^{17,23} PolyP also alters the viability of kidney cells although less is known about the mechanism.¹⁹

HRG (histidine-rich glycoprotein) is a 75-kDa glycoprotein that is synthesized in the liver and that circulates in plasma at 1 to 3 μmol/L.²⁴ HRG modulates the prothrombotic effect of intraperitoneal polyP injection in mice because compared with wild-type mice, HRG-deficient (HRG^{-/-}) mice exhibit increased thrombin generation and fibrin deposition, effects abolished with HRG supplementation.²⁵ There are at least 2 mechanisms by which HRG attenuates the prothrombotic effects of intraperitoneal polyP. First, HRG binds polyP and neutralizes its procoagulant activities.²⁵ Second, HRG binds FXIIa (activated FXII) and attenuates its capacity to promote autoactivation and to activate FXI.²⁵⁻²⁷ The relative importance of these 2 mechanisms is uncertain.

When given by intravenous injection to mice, concentrations of polyP up to 300 mg/kg cause death within minutes.^{7,28,29} Mortality was attributed to overwhelming thrombosis, and consequently, high-dose intravenous polyP infusion is often used as a trigger of thrombosis in animal models. However, neither the protective effect of HRG nor the potential cardiotoxicity of polyP has been considered. Because intraperitoneal polyP induces thrombosis that is exaggerated with HRG depletion and attenuated with FXII depletion, we set out to determine whether HRG and FXII depletion have the same opposing effects in the intravenous polyP model. To explore the protective effects of HRG, we compared the survival of wild-type and HRG^{-/-} mice given increasing intravenous doses of synthetic or platelet-derived polyP. The contribution of thrombosis to the lethal effects of intravenous polyP was investigated by determining whether

Highlights
• With high doses, intravenous polyphosphate is lethal in mice. • Lethality reflects cardiotoxicity rather than thrombosis. • Histidine-rich glycoprotein protects against the lethal effects of polyphosphate.

pretreatment with enoxaparin or knockdown of FXII with an antisense oligonucleotide (ASO) rescued the mice and attenuated thrombosis. The potential cardiotoxicity of polyP was investigated by continuous monitoring of the ECG, by quantification of the levels of troponin I and the myocardial band of creatine kinase (CK-MB) before and after polyP injection, and by assessment of the effects of polyP on the viability of a cultured murine cardiomyocyte cell line in the absence or presence of HRG. Finally, the potential toxic effect of polyP on the kidneys was investigated by measuring the creatinine levels before and after polyP injection and by examining the effect of polyP on the viability of cultured HEK (human embryonic kidney) cells in the absence or presence of HRG.

METHODS

The authors declare that all supporting data are available within the article (and the [Supplemental Material](#)).

Reagents and Isolation of Platelet-Derived polyP

The source of reagents and the methods for the isolation of platelet-derived polyP, in vivo studies, histology, and cell culture are available in the Major Resources Table in the [Supplemental Material](#). Synthetic short-chain polyP, which was generously provided by Dr Thomas Staffel (70.5 CAREPHOS 244; BK Giulini, GmbH, Ladenburg, Germany), was dissolved in 20 mmol/L HEPES, pH 7.4, for the in vitro studies and in 0.9% sterile saline for the in vivo studies without added calcium. PolyP was isolated from platelets as described in the [Supplemental Material](#). The various polyP preparations were subjected to TBE (tris-boric acid)-urea gel analysis and their capacities to promote FXII autoactivation were compared. On TBE-urea gel analysis, the size distribution of platelet-derived polyP was like that of synthetic short-chain polyP from BK Giulini and commercially available polyP-100 from Kerafast (Boston, MA), whereas longer polyP polymers were evident in synthetic long-chain polyP from BK Giulini ([Figure S1A](#)). All the polyP preparations promoted FXII autoactivation to a similar extent ([Figure S1B](#)). PolyP concentrations used in this study are given in terms of phosphate monomer (NaPO₃) concentration as described previously.^{3,6}

Mice

HRG^{-/-} mice were maintained on a pure C57BL/6N background (The Jackson Laboratory, Bar Harbor, MA) after >10

generations of backcrossing.^{25,30,31} Wild-type C57BL/6J mice were purchased from The Jackson Laboratory (Bar Harbor, MA) and were bred in-house for at least 5 generations. Mice were housed in Micro-Isolator cages exposed to a 12-hour constant light-dark cycle and were given free access to food and water. Weight- and age-matched female mice were used for all the experiments. Blood was collected, and serum and plasma were prepared as described in the [Supplemental Material](#). All animal utilization protocols were approved by the Animal Research Ethics Board at McMaster University, and studies were performed according to the Canadian Council on Animal Care guidelines.

Intravenous polyP Injection in Mice

HRG^{-/-} and wild-type mice were anesthetized using a combination of oxygen (0.5 L/min) and isoflurane (1%–4%) delivered by face mask. After the skin was aseptically prepared, the jugular vein was exposed via a midline incision in the neck and a cannulus (polyethylene tubing) was inserted and secured in place with a suture. Synthetic or platelet-derived polyP (10–50 mg/kg in saline) or an equivalent volume of saline was given intravenously via an infusion pump (Harvard Apparatus, Holliston, MA) at a rate of 2 μ L/min per g. Both the synthetic and platelet-derived polyP preparations tested negative for endotoxin activity in a screening assay (Pierce Endotoxin Chromogenic Kit from Thermo Fisher). For comparison purposes, an intravenous dose of 50 mg/kg polyP produces a circulating polyP concentration of about 7 mmol/L assuming a 1-mL blood volume for a 25-g mouse. The plasma and tissue samples were collected 30 minutes after treatment with either saline or polyP or at the time of death if it occurred earlier. Mice were euthanized via cervical dislocation, blood was collected via puncture of the inferior vena cava, and plasma or serum was prepared as detailed in the [Supplemental Material](#) (Figure S2). The lungs, liver, heart, and kidneys were removed for histological analysis.

Effects of HRG Supplementation, FXII Knockdown, or Treatment With Enoxaparin

To investigate the potential protective effects of HRG and the contribution of FXII in this model, some HRG^{-/-} mice were given a retro-orbital injection of 6 mg/kg of human HRG before the intravenous administration of 30 mg/kg polyP. This dose of supplemental human HRG was chosen because it raises the plasma HRG concentration to 1.5 to 2 μ mol/L, which is similar to that in wild-type mice.³¹

To determine whether thrombosis was responsible for the toxicity of polyP, we examined the effect of FXII knockdown or enoxaparin. The levels of FXII in wild-type or HRG^{-/-} mice were knocked down with a liver-directed, mouse-specific FXII-directed ASO generously provided by Ionis Pharmaceuticals (Carlsbad, CA).²⁵ The FXII ASO was administered subcutaneously at a dose of 20 mg/kg twice weekly for 4 weeks (Figure S3), and the extent of knockdown was assessed by comparing hepatic mRNA and plasma FXII levels (Figure S4) before and after ASO administration as described previously.²⁵ Treatment with FXII ASO significantly reduced FXII levels by over 99% ($P < 0.001$; Figure S4). Enoxaparin (10 mg/kg) was given via the retro-orbital vein 30 minutes before the intravenous

injection of 50 mg/kg polyP or an equivalent volume of saline. This dose of enoxaparin produced plasma anti-Xa and anti-IIa levels of 1.2 ± 0.2 and 0.5 ± 0.1 U/mL, respectively (Figure S5), consistent with the reported anti-Xa/anti-IIa ratio of enoxaparin of about 3 to 1.^{32,33} The anti-Xa level was within the reported therapeutic range of 0.6 to 1.0 IU/mL.^{32,33}

ECG Monitoring

To assess for cardiotoxicity, the ECG was continuously monitored after induction of anesthesia using an animal BioAmp recorder (AD Instruments) and subdermal electrodes (AD Instruments, Sydney, Australia) inserted into the right hind foot, left chest/breast, and right neck-shoulder area. ECG tracings were recorded starting 2 minutes before jugular vein cannulation and continuing until 30 minutes after polyP or saline injection or until death. The ECG tracings were analyzed by 3 investigators (R.A.M., J.Z., and J.S.H., an electrophysiologist), who were blinded to treatment allocation, using a combination of visual inspection and LabChart 8 software (AD Instruments, Dunedin, New Zealand).

Levels of Biomarkers

Serum levels of murine troponin I and CK-MB were quantified using immunoassays (Biomatik, Cambridge, Ontario, Canada) according to the manufacturer's instructions. Plasma creatinine levels were quantified using a kit purchased from Crystal Chem (Elk Grove Village, IL). Plasma thrombin-antithrombin (TAT) complex levels were quantified by immunoassay (Affinity Biologicals, Ancaster, Ontario, Canada) as described previously.²⁵

Effects of polyP on Cultured Cells

Mouse cardiomyocyte HL-1 and HEK293 cell line were cultured to about 60% confluence in 8-well chambers coated with fibronectin (10 μ g/mL per well). The cells were exposed to polyP in concentrations up to 42 mmol/L or buffer for up to 24 hours. Cell cytotoxicity was assessed by measuring the levels of LDH (lactate dehydrogenase) in the culture medium using a commercial kit (Thermo Fisher, Grand Island, NY). The effect of polyP on cell structure was assessed by staining for actin filaments (F-actin) after the cells were fixed with 4% paraformaldehyde for 30 minutes, permeabilized with 0.1% Triton X-100 in PBS, and blocked by incubation with 1% BSA in PBS for 1 hour at room temperature. Actin filaments were stained with 1X Phalloidin-iFluor-488 (Cayman Chemicals, Ann Arbor, MI), and the nuclei were stained with DAPI (4',6-diamidino-2-phenylindole). The cells were imaged using the Olympus BX41 fluorescence microscope equipped with a DP72 camera (Center Valley, PA), and images were assessed using the ImageJ software, version 1.48 (National Institutes of Health, Bethesda, MD).

Statistical Analysis

All results are presented as mean $\pm$ SD. Unless otherwise stated, experiments were performed at least 3 $\times$. The Shapiro-Wilk test was used to test for normality, and the significance of differences was determined by 1- or 2-way ANOVA followed by the Holm-Sidak or Tukey comparison test. SigmaPlot, version

11 (Systat Software, San Jose, CA), and GraphPad Prism 9 (GraphPad Software, San Diego, CA) were used to create graphs, and IBM SPSS Statistics software (Armonk, NY) was used to perform the statistical analyses. For all analyses, P values <0.05 were considered statistically significant.

RESULTS

Intravenous polyP Is Lethal in Wild-Type and HRG^{-/-} Mice

Wild-type or HRG^{-/-} mice were given intravenous synthetic polyP in doses up to 50 mg/kg or an equivalent volume of saline. With a synthetic polyP dose of 50 mg/kg, all the wild-type and HRG^{-/-} mice died within 5 minutes, whereas all the saline control mice survived (Figure 1A and 1B). With a synthetic polyP dose of 30 mg/kg, 7 of 8 wild-type mice survived, whereas all the HRG^{-/-} mice died with a significantly ($P<0.001$) shorter mean survival time of 5 ± 2 minutes. At 50 mg/kg, platelet-derived and synthetic polyP had similar effects on survival times (Figure S6). Consequently, synthetic polyP was used for subsequent studies.

Procoagulant and Prothrombotic Effects of polyP Are Greater in HRG^{-/-} Mice Than in Wild-Type Mice

Blood was collected from wild-type and HRG^{-/-} mice before and 30 minutes after they were given 50 mg/kg of polyP or saline or at the time of death, whichever came sooner, and plasma recalcification times and TAT levels were determined (Figure 1C). Compared with saline, 50 mg/kg of polyP shortened the recalcification time in a dose-dependent manner (Figure 1C) up to 2.7-fold in wild-type mice (from 898 ± 58 to 394 ± 37 s; $P<0.001$) and up to 2.5-fold in HRG^{-/-} mice (from 716 ± 78 to 289 ± 29 s; $P<0.0001$). PolyP increased TAT levels in a dose-dependent manner (Figure 1D), and the levels were significantly higher in HRG^{-/-} mice than in wild-type mice ($P<0.001$). Compared with saline, polyP increased the TAT levels up to 10-fold in wild-type mice (from 4 ± 0.6 to 50 ± 6 ng/mL; $P<0.0001$) and up to 14-fold in HRG^{-/-} mice (from 8 ± 1 to 71 ± 8 ng/mL; $P<0.0001$). Likewise, polyP induced dose-dependent fibrin deposition in the lungs, which was quantified by immunofluorescence using a fibrin-directed antibody (Figure 1E and 1F). Fibrin deposition was greater in HRG^{-/-} mice than in wild-type mice ($P=0.005$). For example, compared with saline-treated mice, treatment with 50 mg/kg of polyP enhanced fibrin deposition by $10\pm4\%$ ($P<0.0001$) in wild-type mice and by $18\pm4\%$ ($P<0.0001$) in HRG^{-/-} mice, a statistically significant difference ($P=0.005$). Therefore, the procoagulant and prothrombotic effects of polyP are greater in HRG^{-/-} mice than in wild-type mice.

To determine whether thrombosis is responsible for the toxicity of polyP, mice were injected with 10 mg/kg of enoxaparin or saline 30 minutes before the administration of 50 mg/kg of polyP. Alternatively, 50 mg/kg of polyP was given to mice that were pretreated for 4 weeks with an FXII-directed or a control ASO (Figure S3). Although FXII depletion and enoxaparin significantly prolonged the recalcification clotting times ($P<0.001$), reduced the TAT levels ($P<0.001$; Figure 2C and 2D), and attenuated fibrin deposition in the lungs ($P<0.0001$; Figure 2E and 2F), neither intervention improved survival (Figure 2A and 2B). These findings suggest that overwhelming thrombosis is not the major driver of the lethal effects of polyP. This concept was supported by our failure to detect thrombosis in the coronary, cerebral, or main pulmonary arteries, which could cause death in minutes.

Intravenous polyP Induces ECG Changes

To assess the potential cardiotoxicity of polyP, the ECG was continuously monitored. Neither anesthesia nor surgery was associated with changes in the ECG (Figure 3A and 3B). However, injection of synthetic or platelet-derived polyP induced dose-dependent ECG changes, including conduction defects (premature ventricular contractions and first-, second-, or third-degree atrioventricular block) and ST-segment elevation or depression (Figure 3C through 3H). The ECG abnormalities were more marked in HRG^{-/-} mice than in wild-type mice (Figure 3I through 3K). For example, 10 mg/kg of polyP induced first-degree block in wild-type mice and second-degree block in HRG^{-/-} mice (Figure 3I and 3J). With 30 mg/kg polyP, wild-type mice exhibited first- or second-degree block, whereas in addition to the third-degree block, ST-segment elevation or depression was observed in HRG^{-/-} mice. Neither enoxaparin nor FXII knockdown reversed the ECG changes associated with polyP injection. These observations suggest that polyP has toxic effects on the heart that are unrelated to thrombosis.

PolyP Increases the Levels of Cardiac and Kidney Biomarkers

To further examine the potential cardiotoxicity and kidney toxicity of polyP, serum levels of troponin I and CK-MB were measured as were the creatinine levels. Compared with saline, polyP increased the levels of troponin I and CK-MB in a dose-dependent manner. Troponin I levels were increased by 2.4-fold ($P=0.003$) and 3.2-fold ($P<0.001$) in mice treated with 30 and 50 mg/kg polyP, respectively (Figure 4A). The levels of troponin I were significantly higher in HRG^{-/-} mice than in wild-type mice (Figure 4A and 4B) both with the 30-mg/kg ($P=0.01$) and 50-mg/kg polyP dose ($P<0.001$).

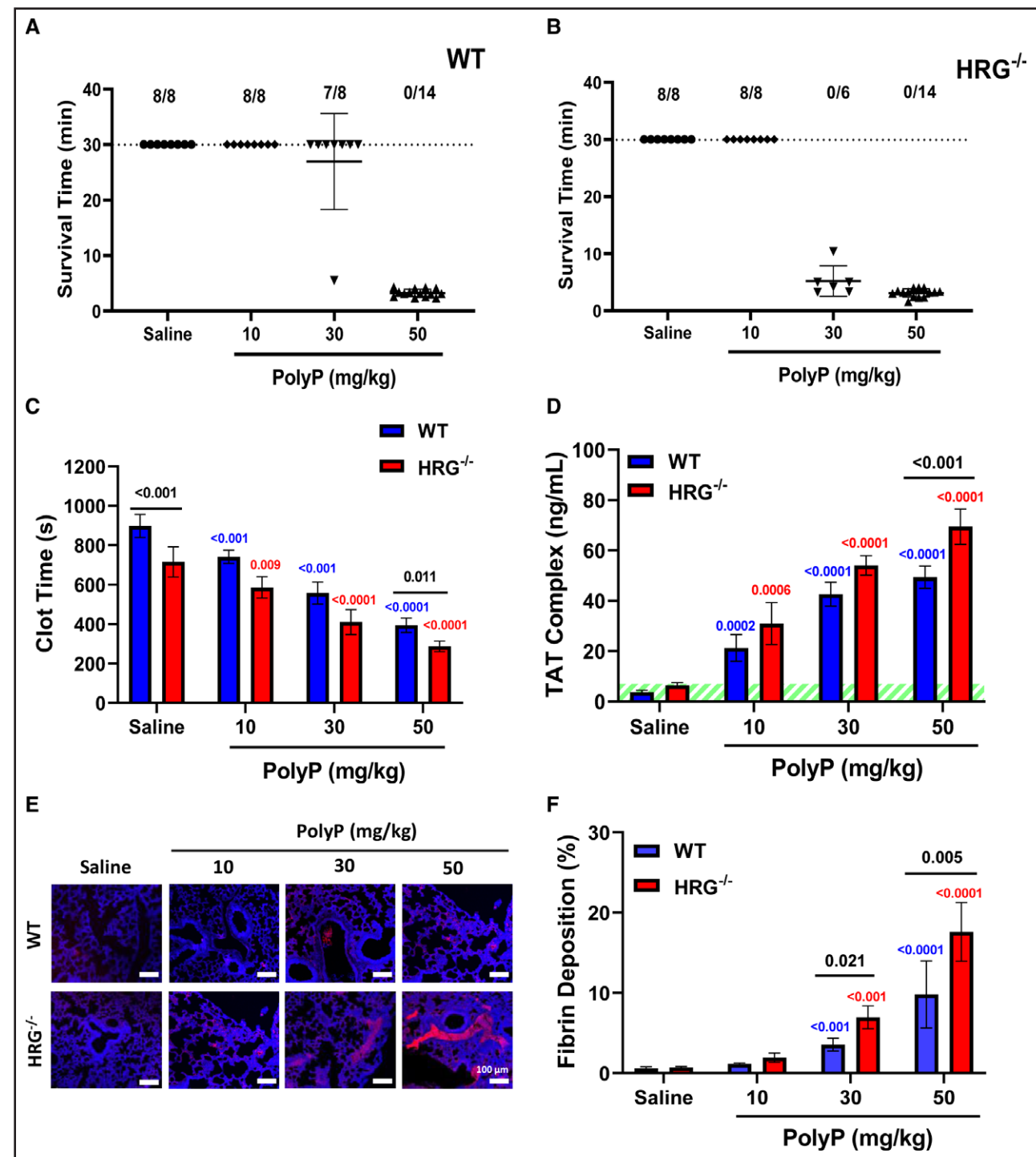


Figure 1. Polyphosphate (PolyP) has dose-dependent effects on survival time, recalcification time, and thrombosis in wild-type (WT) and histidine-rich glycoprotein-deficient (HRG^{-/-}) mice.

Survival time of WT (**A**) or HRG^{-/-} (**B**) mice given intravenous saline or polyP at the doses indicated. **C**, Plasma recalcification times in WT (blue bars) or HRG^{-/-} (red bars) mice given saline or polyP at the doses indicated. **D**, Thrombin-antithrombin (TAT) complex levels in plasma from WT (blue bars) or HRG^{-/-} (red bars) mice. The green hatched area represents normal TAT levels in mice. **E**, Fibrin deposition in the lungs of WT or HRG^{-/-} mice was identified by immunofluorescence using an anti-fibrin(ogen) antibody (red). DAPI (4',6-diamidino-2-phenylindole; blue) was used as a nuclear stain. Representative confocal images are shown, and the scale bars represent 100 μm. **F**, Fibrin staining was quantified as the percentage of the total lung area covered by fibrin staining. $n=6$ to 14 mice per group. Bars represent mean $\pm$ SD. P values reflect comparison with saline-treated WT (blue text) or HRG^{-/-} (red text) mice using 2-way ANOVA (Tukey method).

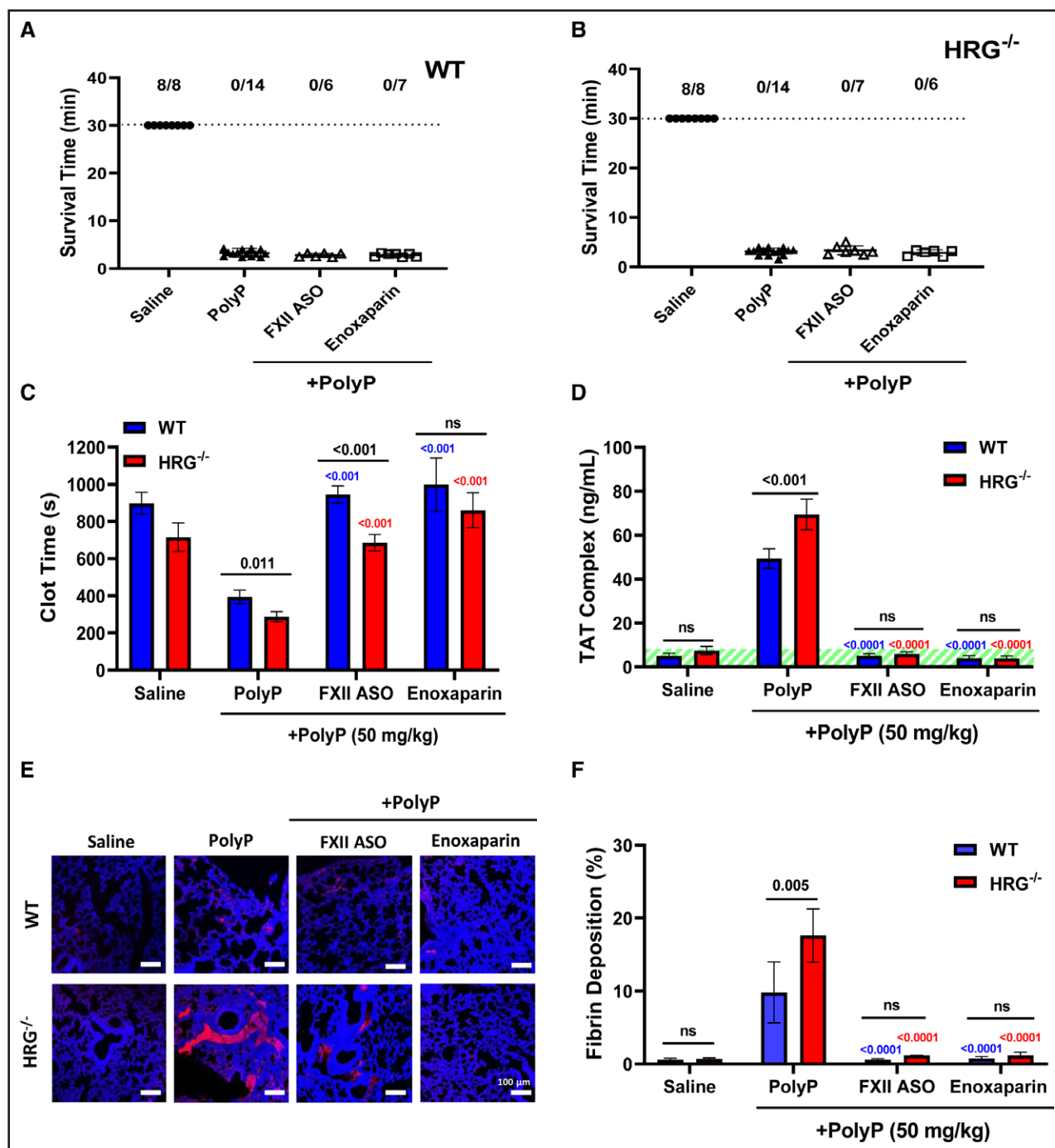


Figure 2. Effect of FXII (factor XII) knockdown or enoxaparin on the procoagulant and prothrombotic effects of polyphosphate (polyP) in wild-type (WT) and histidine-rich glycoprotein-deficient (HRG^{-/-}) mice.

Survival time in WT (A) or HRG^{-/-} (B) mice given intravenous saline, 50 mg/kg of polyP, or pretreated with a FXII antisense oligonucleotide (ASO; 40 mg/kg per week for 4 weeks) or enoxaparin (10 mg/kg) before administration of 50 mg/kg of polyP. C, Plasma recalcification times and (D) thrombin-antithrombin (TAT) complex levels in plasma from WT or HRG^{-/-} mice in A and B. E, Fibrin deposition in the lungs of WT or HRG^{-/-} mice was stained using an anti-fibrin(ogen) antibody (red). DAPI (4',6-diamidino-2-phenylindole) was used as a nuclear stain (blue). F, Images were captured, and fibrin staining was quantified and calculated as the percentage of the total lung area covered by fibrin. Scale bars represent 100 μm. n=6 to 14 mice per group. Bars represent mean±SD. P values reflect comparison with polyP-treated WT (blue text) or HRG^{-/-} (red text) mice only (2-way ANOVA, Tukey method).

To determine whether intravenous polyP also has adverse effects on the kidneys, the plasma levels of creatinine were quantified. PolyP increased plasma creatinine

levels in a dose-dependent manner (Figure 4C), and the creatinine levels in HRG^{-/-} mice were significantly higher than those in wild-type mice ($P<0.001$).

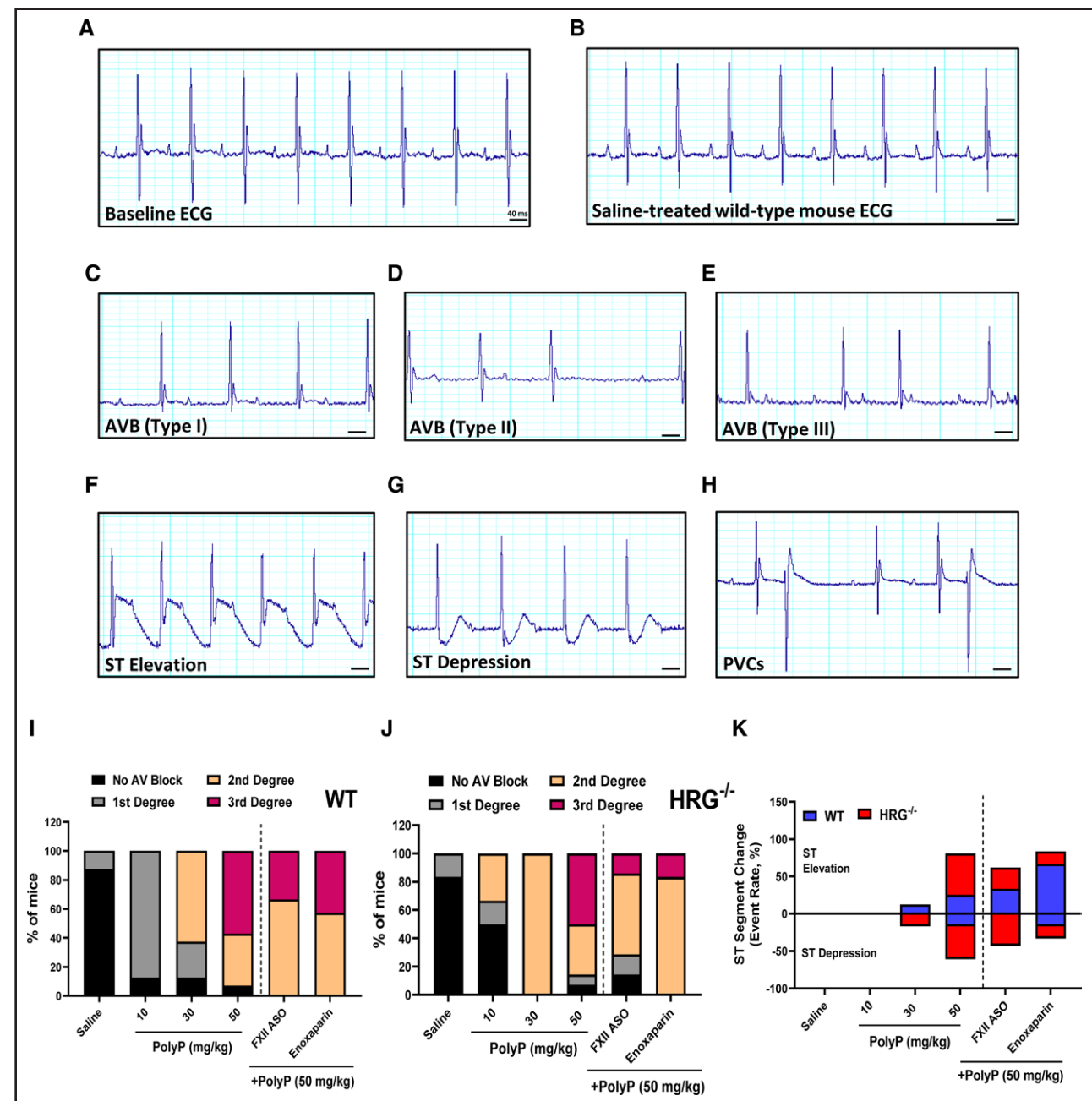


Figure 3. ECG changes associated with polyphosphate (polyP) administration to wild-type (WT) or histidine-rich glycoprotein-deficient (HRG^{-/-}) mice.

A, Representative baseline ECG tracing from a mouse under anesthesia before surgery or any treatment. **B**, Normal ECG tracing 30 minutes after intravenous saline administration. Representative ECG tracings revealing first-, second-, or third-degree atrioventricular block (AVB; **C–E**), ST-segment elevation (**F**), ST-segment depression (**G**), or premature ventricular contractions (PVCs; **H**) after polyP administration. The percentage of WT (**I**) or HRG^{-/-} (**J**) mice with AVB after treatment with polyP at the doses shown and the effect of FXII (factor XII) knockdown or enoxaparin on AVB in mice given 50 mg/kg of polyP. **K**, ST-segment changes (elevation or depression) in WT (blue bars) or HRG^{-/-} (red bars) mice after treatment with polyP at the doses shown and the effect of FXII knockdown or enoxaparin on AVB in mice given 50 mg/kg of polyP. ASO indicates antisense oligonucleotide; and AV, atrioventricular.

Neither enoxaparin nor FXII knockdown reduced the levels of troponin I, CK-MB, or creatinine to control levels in wild-type or HRG^{-/-} mice (Figure 4D through 4F). These results suggest that the organ damage associated with polyP administration is not mediated by thrombosis.

HRG Supplementation Protects HRG^{-/-} Mice From the Lethal Effect of polyP

To confirm that HRG is protective, HRG^{-/-} mice were given 6 mg/kg of human HRG or saline before injection of 30 mg/kg of polyP. None of the mice given human

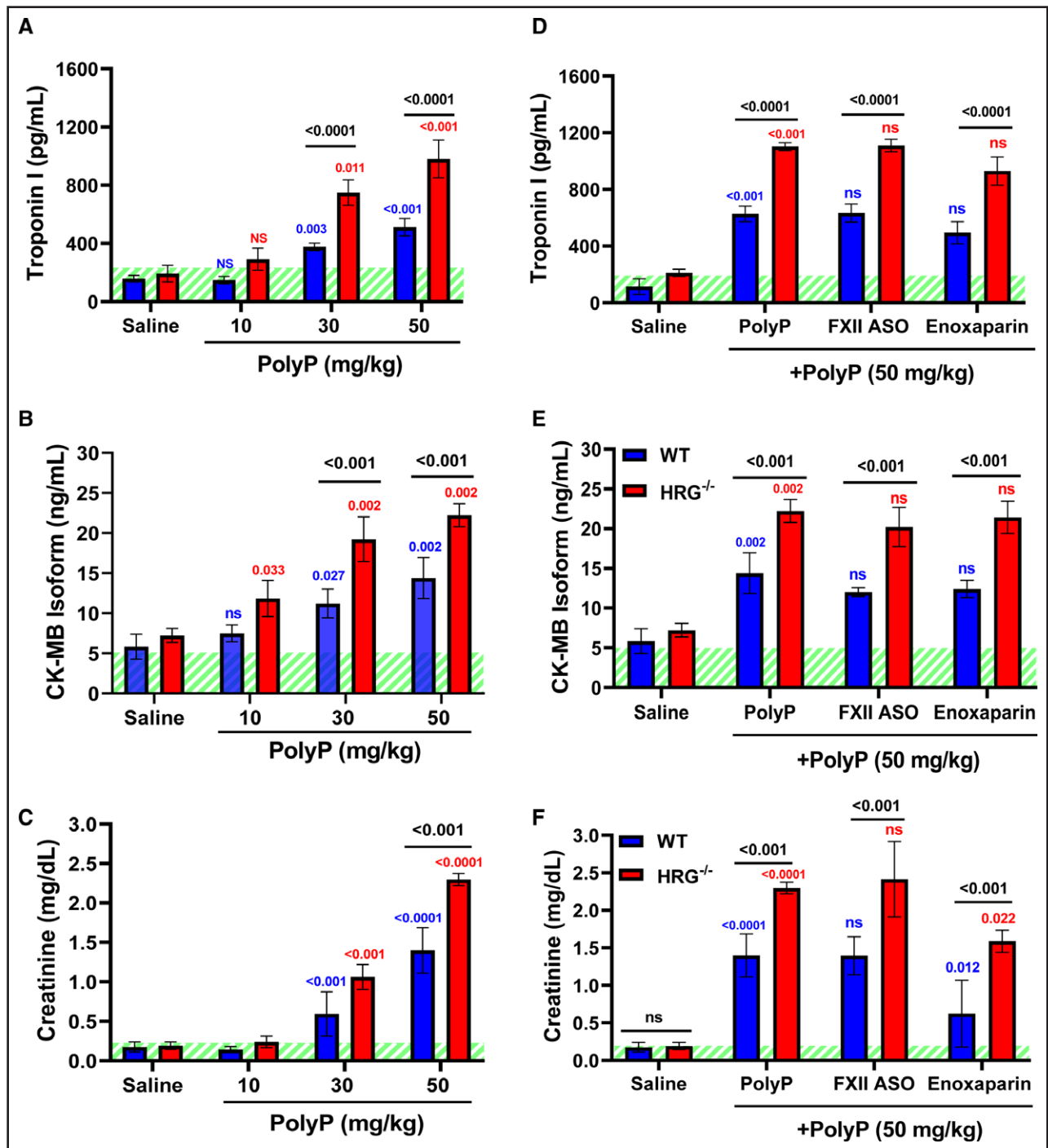


Figure 4. Effect of polyphosphate (polyP) on the levels of troponin I, creatine kinase myocardial band (CK-MB), and creatinine in wild-type (WT) and histidine-rich glycoprotein-deficient (HRG^{-/-}) mice.

A through C, Troponin I (**A**), CK-MB (**B**), and creatinine (**C**) levels in WT (blue bars) or HRG^{-/-} (red bars) mice treated with saline or polyP at the doses shown. **D through F,** Troponin I (**D**), CK-MB (**E**), and creatinine (**F**) levels in WT (blue bars) or HRG^{-/-} (red bars) mice given 50 mg/kg polyP with or without FXII (factor XII) knockdown or enoxaparin pretreatment. The green hatched areas represent the normal ranges for the biomarkers. *n*=6 to 14 mice per treatment group. Bars represent mean±SD. *P* values reflect comparisons with saline-treated mice in mice given polyP alone. For mice pretreated with the FXII antisense oligonucleotide (ASO) or given enoxaparin, *P* values reflect comparisons with WT (blue text) or HRG^{-/-} (red text) mice given 50 mg/kg polyP (2-way ANOVA, Tukey method).

HRG died, whereas all of those given saline died (Figure 5A). The mean plasma HRG level in the HRG^{-/-} mice supplemented with human HRG was 136±7 µg/mL,

which was comparable to the mean level of 127±12 µg/mL in wild-type mice (Figure 5B). HRG supplementation not only significantly reduced fibrin deposition in the

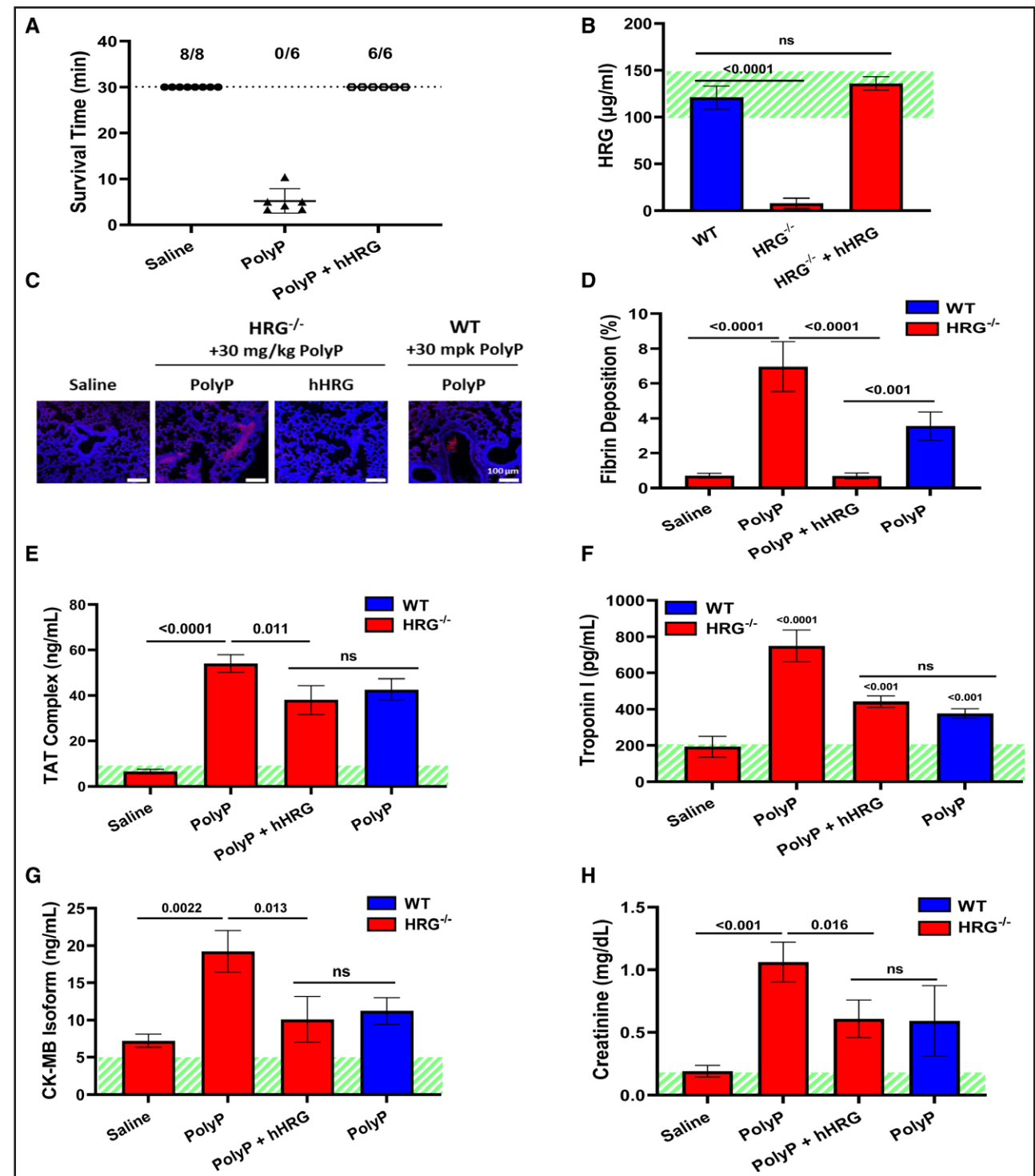


Figure 5. Effect of HRG (histidine-rich glycoprotein) supplementation on thrombosis and levels of biomarkers in HRG-deficient (HRG^{-/-}) mice given 30 mg/kg of polyphosphate (polyP).

A, Survival times of HRG^{-/-} mice given saline or 30 mg/kg of polyP without or with pretreatment with 6 mg/kg HRG. **B**, Plasma HRG levels in wild-type (WT) mice (blue bar) and HRG^{-/-} mice without or with HRG supplementation (red bars). **C** and **D**, Fibrin deposition and quantification in the lungs of WT mice (blue bars) or HRG^{-/-} mice given saline or 30 mg/kg of polyP without or with supplemental HRG (red bars). Scale bars represent 100 μm. Thrombin-antithrombin (TAT) complex (**E**), troponin I (**F**), creatine kinase myocardial band (CK-MB; **G**), and creatinine (**H**) levels in WT (blue bars) or HRG^{-/-} mice (red bars) given saline or 30 mg/kg of polyP without or with supplemental HRG. The green hatched areas represent the normal ranges of biomarkers. n=6 to 8 mice per group. Bars represent mean±SD. *P* values reflect 2-way ANOVA (Tukey method). hHRG indicates human histidine-rich glycoprotein.

lungs ($P<0.0001$; Figure 5C and 5D) but also attenuated the ECG abnormalities induced by polyP (Figure S8) and significantly reduced the levels of TAT ($P=0.011$), troponin I ($P=0.011$), CK-MB ($P=0.013$), and creatinine ($P=0.016$) to levels comparable with those in wild-type mice (Figure 5E through 5H). Therefore, HRG appears to protect against both the thrombotic effects of polyP and its toxic effects on the heart and kidneys.

PolyP Is Toxic for Cultured Cells and HRG Is Protective

To evaluate the cellular toxicity of polyP and the protective effects of HRG, a cultured murine cardiomyocyte cell line (HL-1) and cultured HEK293 cells were exposed to polyP in the absence or presence of HRG. The doses of polyP were chosen to be comparable with the levels achieved in mice given intravenous polyP. Exposure of HL-1 cells to polyP resulted in a time- and dose-dependent increase in the levels of LDH in the culture

medium (Figure 6A; Figure S9). The increase in LDH levels likely reflects cell damage because there was an associated increase in F-actin staining (Figure 6B). Furthermore, 7 mmol/L polyP, a concentration similar to that achieved in mice given a 50-mg/kg dose of polyP, disrupted the cell cytoskeleton in a time-dependent manner and caused blebbing of the cell membrane, cell shrinkage, and nuclear condensation. In cells exposed to polyP, HRG significantly ($P=0.009$) decreased LDH levels in a concentration-dependent manner and reduced F-actin staining (Figure 6C and 6D). PolyP had similar effects on HEK293 cells (Figure S9). Therefore, polyP is toxic for both HL-1 and HEK293 cells and HRG protects against its toxic effects.

Alkaline Phosphatase Digestion Eliminates the Toxic Effects of polyP

To ensure the polyP was responsible for the toxic effects in mice, its effects after alkaline phosphate digestion

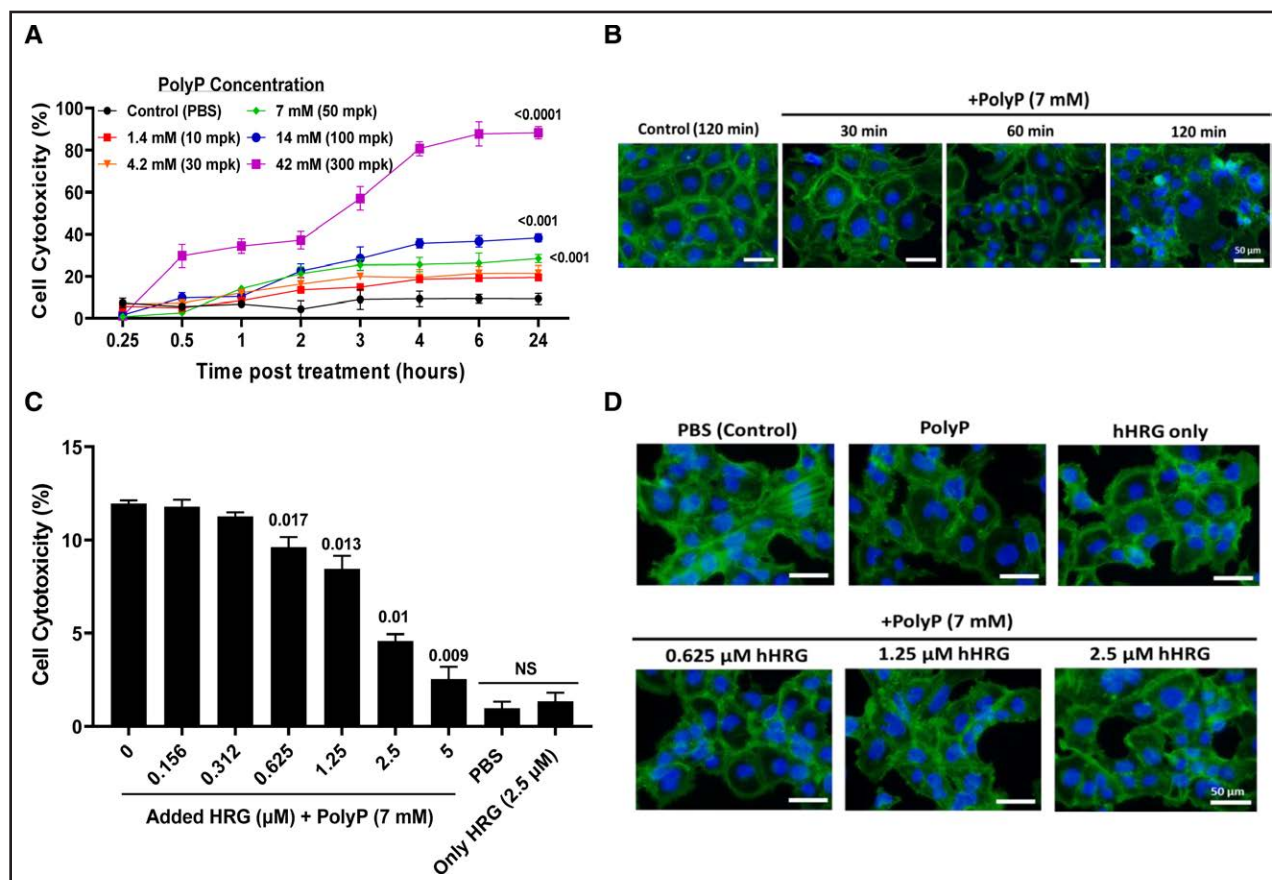


Figure 6. Effects of polyphosphate (polyP) on HL-1 cells in culture.

A and **C**, PolyP, at the concentrations shown, was added to confluent HL-1 cells in culture, and LDH (lactate dehydrogenase) levels in the medium were quantified at the indicated time points. LDH levels were converted to percent cell cytotoxicity using a standard curve. The culture media was supplemented with up to 2.5 μmol/L HRG (histidine-rich glycoprotein), and after 60-minute incubation with 7 mmol/L polyP, cell cytotoxicity was quantified. $n=5$ determinations performed in duplicate per treatment group; mean $\pm$ SD values are shown. P values were determined by performing a 1-way ANOVA (Holm-Sidak method). **B** and **D**, The effect of HRG supplementation on cell structure was assessed by staining actin filaments with Phalloidin F-actin-488 (green) and nuclei with DAPI (4',6-diamidino-2-phenylindole; blue). Images were visualized using a $\times 40$ objective lens and an Olympus BX41 microscope equipped with a DP72 camera (Olympus, Tokyo, Japan). Scale bars represent 50 μm. P values represent 1-way ANOVA (Holm-Sidak method). hHRG indicates human histidine-rich glycoprotein.

were investigated (Figure S10). SDS-PAGE analysis confirmed polyP digestion. Wild-type and HRG^{-/-} mice were given 50 mg/kg of digested polyP or untreated polyP. All the mice survived after digested polyP injection, whereas none survived after untreated polyP injection. Furthermore, with alkaline phosphatase-digested polyP, there was no significant fibrin deposition in the lungs or increase in the levels of TAT, troponin I, CK-MB, or creatinine.

DISCUSSION

Prior studies have shown that when administered to mice intravenously in high doses, both short-chain and long-chain polyP cause death within minutes, which has been attributed to overwhelming thrombosis.^{7,28,29,34} We have previously shown that HRG protects mice from thrombosis induced by intraperitoneal injection of short-chain polyP by binding polyP and neutralizing its activity or binding FXIIa and attenuating its procoagulant activity.²⁵ The current study was undertaken to determine whether HRG and FXII have the same opposing effects with polyP in a murine model wherein polyP is given intravenously in high doses. We show that (1) at high doses, intravenous short-chain polyP is lethal in mice; (2) death is not due to thrombosis because neither enoxaparin nor FXII depletion enhances survival; (3) death likely reflects cardiotoxicity because polyP induces ECG abnormalities, increases the levels of troponin I and CK-MB in a dose-dependent manner, and is toxic for cultured HL-1 cells; and (4) HRG protects against the thrombotic and cardiotoxic effects of polyP. Therefore, HRG modulates both the prothrombotic effects of polyP and its toxic effects on cells.

Previous studies have shown that intravenous polyP is lethal when given to mice in doses up to 300 mg/kg.^{7,28,29,34} In our study, the lethal dose of polyP in wild-type mice was ≥ 50 mg/kg, which corresponds to blood levels of at least 1 mg/mL. This amount is equivalent to a phosphate monomer concentration of about 7 mmol/L. To put this into perspective, the concentration of polyP released from activated platelets in whole blood is estimated to reach 2 to 7 μ mol/L and possibly higher in the vicinity of platelet-rich thrombi.⁶ Although the doses of polyP used in our study are higher than the physiological levels of polyP, we used these high doses to investigate whether HRG modulates the toxic effects of polyP and to determine whether mortality is due to thrombosis, as per the current dogma, or due to cardiotoxicity.

FXII knockdown or enoxaparin reduced TAT levels and attenuated fibrin deposition in the lungs indicating protection against the thrombotic effects of polyP. However, neither of the anticoagulant strategies protected against the cardiotoxic effects of polyP, nor did they prevent the increase in creatinine levels. These findings suggest that the toxic effect of polyP on cells is distinct from its

prothrombotic effects. That FXII knockdown failed to protect mice from the lethal effects of polyP is at odds with a previous report that FXII^{-/-} mice are protected against the lethal effect of high-dose intravenous polyP.⁷ These divergent findings may reflect incomplete FXII depletion with the dose of the FXII-directed ASO used in our study or a unique feature of the FXII^{-/-} strain of mice used in the previous study. Regardless of the divergent effects of high-dose polyP on mortality with FXII deficiency and knockdown, we show that, like FXII knockdown, enoxaparin attenuates the prothrombotic effects of polyP but does not affect mortality. Therefore, our findings suggest that the lethal effect of polyP is not the result of overwhelming thrombosis.

Supplementation with HRG rescued HRG^{-/-} mice from the prothrombotic and cardiotoxic effects of 30 mg/kg of polyP, suggesting that HRG modulates the cardiotoxicity of polyP via a mechanism distinct from its capacity to bind FXIIa and attenuate its procoagulant activity. Thus, it is likely that HRG neutralizes polyP *in vivo*. This concept is supported by our previous observation that with intraperitoneal injection of polyP, the plasma polyP concentration was 2-fold higher in HRG^{-/-} mice than in wild-type mice.²⁵ Previously, we showed that HRG binds polyP and attenuates its capacity to activate FXII or to promote FXIIa-mediated FXI activation.^{25,26} The results of the current study suggest that by binding polyP, HRG also attenuates its cardiotoxic effects in mice. Therefore, HRG modulates both the prothrombotic and cytotoxic effects of polyP.

The findings that polyP increases the levels of troponin, CK-MB, and creatinine and impairs the viability of HL-1 cells, a murine cardiomyocyte cell line, and HEK293 cells, a human kidney cell line, in a dose-dependent manner suggest that the cytotoxic effects of polyP are not cell or organ specific. This observation is consistent with previous reports that polyP induces damage in neurocytes and various blood cells, as well as cardiomyocytes.^{35,36} In cardiomyocytes, polyP activates the mitochondrial permeability transition pore allowing the influx of ions, including calcium.^{21,36,37} PolyP binds calcium to form apatite, a calcium-phosphate complex.^{17,20,38–40} Apatite adversely affects mitochondrial permeability transition pore homeostasis and increases the frequency of mitochondrial permeability transition pore opening.^{21,36,37} By binding polyP, HRG has been shown to attenuate apatite formation,^{39,40} which may explain why HRG protects cultured HL-1 cells from the toxic effects of polyP.

A potential limitation of this study is the use of wild-type mice as controls instead of HRG^{+/+} littermates. Although differences between the wild-type C57BL/6J and the HRG^{-/-}C57BL/6 N strains have been reported,⁴¹ these are unlikely to have influenced our findings because (1) to the best of our knowledge, there are no reported differences in pathways related to thrombosis, (2) the studies lasted ≤ 30 minutes, a time frame that is too short to

be influenced by strain-related differences, (3) the effect of polyP and saline was compared in both strains, and (4) supplemental HRG rendered the results in HRG^{+/+} mice similar to those in wild-type mice. Therefore, the use of wild-type mice as controls instead of HRG^{+/+} littermates is unlikely to be a drawback in this study.

In summary, the results of this study suggest that intravenous polyP has numerous physiological effects. At low doses, polyP causes thrombosis, whereas at higher doses, it has cytotoxic effects that lead to organ dysfunction. By binding polyP, HRG modulates both phenomena, thereby adding to the repertoire by which HRG down-regulates coagulation.

ARTICLE INFORMATION

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Disclosures

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Supplemental Material

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